

CAPSTONE PROJECT

Health Care Project Insurance Cost

Contents

Topic	Page No.
Introduction	1
Data Report	1 to 4
Exploratory Data Analysis	5 to 11
Business insights from EDA	12 to 14
EDA (Exploratory Data Analysis) and Business Implications	15
Data Cleaning and Pre-processing	15 to 16
Model building and interpretation	17 to 20
Model Tuning	20 to 22
Why was a particular model chosen?	22
Efforts to Improve Model Performance	22 to 23
Final interpretation / Recommendation	23

Lists of Tables

Table	Topic	Page No.
1	Data Dictionary	1
2	Top 5 rows of the dataset	2
3	Last 5 rows of the dataset	2
4	Basic Info of the dataset	3
5	Missing value of the dataset	3
6	Statistical Summary of the dataset	4
7	Dataset after encoding the Nominal and Ordinal columns	16
8	Head of the X_train scaled	17
9	Model Comparison	20

Lists of Figures

Figure	Topic	Page No.
1	Boxplots for all numerical variables	5
2	Distributions of years of insurance with us	5
3	Distribution of regular checkup lasy year	5
4	Distribution of visited doctor last 1 year	6
5	Distribution of cholesterol level	6
6	Distributions of daily avg steps	6
7	Distribution of Age	6
8	Distribution of avg glucose level	6
9	Distribution of BMI	6
10	Distribution of Weight	7
11	Distribution of weight change in last one year	7
12	Distribution of fat percentage	7
13	Distribution of insurance cost	7
14	distribution of adventure sports	8
15	Distribution of Occupation	8
16	Distribution of heart decs history	8
17	Distribution of other major decs history	8
18	Distribution of gender	8
19	Distribution of smoking status	8

20	Distribution of location	9
21	Distribution of covered by any other company	9
22	Distribution of alcohol	9
23	Distribution of exercise	9
24	Insurance cost vs cholesterol level, bmi, age, weight	10
25	Relationship Between Exercise and Insurance Cost	10
26	Impact of Alcohol Consumption on Insurance Cost	11
27	Insurance cost by heart disease and smoking status	11
28	Clustering Analysis on Age and Insurance cost	13
29	Insurance charges by smoking status	14
30	Insurance charges by region	14
31	Feature Importance (Random Forest)	18
32	Actual vs Predicted Insurance Cost (Random Forest)	19
33	Model Comparison (R2 Score)	20
34	Actual vs Predicted Insurance Cost (Tuned Random Forest)	21

1. Introduction

Healthcare costs are rising, making health insurance essential for most people. However, it can be challenging for insurers to set fair premiums that reflect each person's health and lifestyle. This creates an opportunity to build a simple, data-driven model to estimate insurance costs accurately and fairly.

Such a model helps achieve two goals: making healthcare coverage affordable for individuals and helping insurers manage their financial risks. By considering factors like medical history and lifestyle, it can protect people from unexpected expenses and reward healthy habits with lower premiums. Ultimately, this project aims to create a fair, sustainable healthcare system and promote healthier communities.

➤ Need for the Study/Project:

Medical treatment can be very expensive, especially for those without insurance. A fair and accurate model to estimate insurance costs would benefit both individuals and insurance companies. It allows individuals to get a premium based on their specific health and lifestyle, and helps insurers manage risks by considering factors like medical history and habits. This not only protects people from unexpected costs but also encourages healthier choices by offering lower premiums for healthier lifestyles.

➤ Scope of the Project:

This project aims to create a simple, data-driven model to estimate health insurance premiums based on a person's health and lifestyle. The model will use data like medical history, habits, age, and other factors to calculate costs accurately.

Key areas include:

- Personalized Premiums: Setting fair and transparent insurance costs based on individual health and lifestyle.
- Helping Insurers: Giving insurance companies tools to understand health risks and manage costs better.
- Promoting Healthy Habits: Encouraging healthier lifestyles by offering lower premiums to people with low-risk habits.
- Social Benefits: Making health insurance more affordable and fair, helping more people get the coverage they need.

2. Data Report

a) Understanding How Data Was Collected:

The dataset, includes health, lifestyle, and demographic information for people applying for insurance. This data was likely gathered during the application process, with details on health checks, doctor visits, and metrics like BMI and cholesterol. Data collection likely happens yearly, as it covers information from the past year for each individual.

Target Variables: Insurance_Cost

Data dictionary:

Variable	Business Definition
applicant_id	Applicant unique ID
years_of_insurance_with_us	Since how many years customer is taking policy from the same company only
regular_checkup_lasy_year	Number of times customers has done the regular health check up in last one year
adventure_sports	Customer is involved with adventure sports like climbing, diving etc.
Occupation	Occupation of the customer
visited_doctor_last_1_year	Number of times customer has visited doctor in last one year
cholesterol_level	Cholesterol level of the customers while applying for insurance
daily_avg_steps	Average daily steps walked by customers
age	Age of the customer
heart_decs_history	Any past heart diseases
other_major_decs_history	Any past major diseases apart from heart like any operation
Gender	Gender of the customer
avg_glucose_level	Average glucose level of the customer while applying the insurance
bmi	BMI of the customer while applying the insurance
smoking_status	Smoking status of the customer
Year_last_admitted	When customer have been admitted in the hospital last time
Location	Location of the hospital
weight	Weight of the customer
covered_by_any_other_company	Customer is covered from any other insurance company
Alcohol	Alcohol consumption status of the customer
exercise	Regular exercise status of the customer
weight_change_in_last_one_year	How much variation has been seen in the weight of the customer in last year
fat_percentage	Fat percentage of the customer while applying the insurance
insurance_cost	Total Insurance cost

Table1- Data Dictionary

b) Visual inspection of data (rows, columns, descriptive details):

- Top 5 Rows of the dataset

	applicant_id	years_of_insurance_with_us	regular_checkup_lasy_year	adventure_sports	Occupation	visited_doctor_last_1_year	cholesterol_level	daily_avg_steps	age	h
0	5000	3	1	1	Salried	2	125 to 150	4866	28	
1	5001	0	0	0	Student	4	150 to 175	6411	50	
2	5002	1	0	0	Business	4	200 to 225	4509	68	
3	5003	7	4	0	Business	2	175 to 200	6214	51	
4	5004	3	1	0	Student	2	150 to 175	4938	44	

smoking_status	Year_last_admitted	Location	weight	covered_by_any_other_company	Alcohol	exercise	weight_change_in_last_one_year	fat_percentage	insurance_cost
Unknown	NaN	Chennai	67	N	Rare	Moderate	1	25	20978
formerly smoked	NaN	Jaipur	58	N	Rare	Moderate	3	27	6170
formerly smoked	NaN	Jaipur	73	N	Daily	Extreme	0	32	28382
Unknown	NaN	Chennai	71	Y	Rare	No	3	37	27148
never smoked	2004.0	Bangalore	74	N	No	Extreme	0	34	29616

Table2- Top 5 rows of the dataset

- Last 5 Rows of the dataset

	applicant_id	years_of_insurance_with_us	regular_checkup_lasy_year	adventure_sports	Occupation	visited_doctor_last_1_year	cholesterol_level	daily_avg_steps	ag
24995	29995	3	0	0	Salried	4	225 to 250	5614	2
24996	29996	6	0	0	Business	4	200 to 225	4719	5
24997	29997	7	0	1	Student	2	150 to 175	5624	3
24998	29998	1	0	0	Salried	2	225 to 250	10777	2
24999	29999	8	2	0	Business	4	150 to 175	5882	2

smoking_status	Year_last_admitted	Location	weight	covered_by_any_other_company	Alcohol	exercise	weight_change_in_last_one_year	fat_percentage	insurance_cost	
smokes	2000.0	Kanpur	79		Y	Rare	Moderate	4	40	39488
ever smoked	2009.0	Kanpur	66		N	Rare	Moderate	2	28	14808
Unknown	NaN	Bhubaneswar	76		N	Rare	Moderate	1	35	33318
Unknown	NaN	Surat	74		N	Rare	Moderate	0	40	30850
formerly smoked	2014.0	Chennai	57		N	No	No	4	21	6170

Table3- Last 5 rows of the

>Observations; The dataset contained 25000 rows and 24 columns.

- Basic Info of the dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25000 entries, 0 to 24999
Data columns (total 24 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   applicant_id                          25000 non-null  int64
1   years_of_insurance_with_us            25000 non-null  int64
2   regular_checkup_lasy_year              25000 non-null  int64
3   adventure_sports                       25000 non-null  int64
4   Occupation                             25000 non-null  object
5   visited_doctor_last_1_year             25000 non-null  int64
6   cholesterol_level                     25000 non-null  object
7   daily_avg_steps                       25000 non-null  int64
8   age                                    25000 non-null  int64
9   heart_decs_history                    25000 non-null  int64
10  other_major_decs_history               25000 non-null  int64
11  Gender                                 25000 non-null  object
12  avg_glucose_level                     25000 non-null  int64
13  bmi                                    24010 non-null  float64
14  smoking_status                        25000 non-null  object
15  Year_last_admitted                     13119 non-null  float64
16  Location                               25000 non-null  object
17  weight                                 25000 non-null  int64
18  covered_by_any_other_company           25000 non-null  object
19  Alcohol                                25000 non-null  object
20  exercise                               25000 non-null  object
21  weight_change_in_last_one_year          25000 non-null  int64
22  fat_percentage                         25000 non-null  int64
23  insurance_cost                         25000 non-null  int64
dtypes: float64(2), int64(14), object(8)
memory usage: 4.6+ MB
```

Table4- Basic Info of the dataset

>Observations; In this dataset there are 2 float64, 14 int64 and 8 object datatypes are present. BMI and Year_last_admitted shows missing values are present in those columns.

- Checking Missing values

```
applicant_id                0
years_of_insurance_with_us  0
regular_checkup_lasy_year    0
adventure_sports             0
Occupation                   0
visited_doctor_last_1_year   0
cholesterol_level            0
daily_avg_steps              0
age                           0
heart_decs_history           0
other_major_decs_history     0
Gender                       0
avg_glucose_level            0
bmi                           990
smoking_status               0
Year_last_admitted           11881
Location                     0
weight                       0
covered_by_any_other_company  0
Alcohol                      0
exercise                     0
weight_change_in_last_one_year 0
fat_percentage               0
insurance_cost               0
dtype: int64
```

Table5- Missing value of the dataset

>Observations; 'bmi' column have present 990 missing values and 'Year_last_admitted' column have present 11881 missing values. We see that the 'applicant_id' and 'Year_last_admitted' are not going to be relevant for the analysis so we dropped these variables. We treat the missing values of the 'bmi' column with its mean value.

- Statistical Summary of the dataset

	count	mean	std	min	25%	50%	75%	max
applicant_id	25000.0	17499.500000	7217.022701	5000.0	11249.75	17499.5	23749.25	29999.0
years_of_insurance_with_us	25000.0	4.089040	2.606612	0.0	2.00	4.0	6.00	8.0
regular_checkup_lasy_year	25000.0	0.773680	1.199449	0.0	0.00	0.0	1.00	5.0
adventure_sports	25000.0	0.081720	0.273943	0.0	0.00	0.0	0.00	1.0
visited_doctor_last_1_year	25000.0	3.104200	1.141663	0.0	2.00	3.0	4.00	12.0
daily_avg_steps	25000.0	5215.889320	1053.179748	2034.0	4543.00	5089.0	5730.00	11255.0
age	25000.0	44.918320	16.107492	16.0	31.00	45.0	59.00	74.0
heart_decs_history	25000.0	0.054640	0.227281	0.0	0.00	0.0	0.00	1.0
other_major_decs_history	25000.0	0.098160	0.297537	0.0	0.00	0.0	0.00	1.0
avg_glucose_level	25000.0	167.530000	62.729712	57.0	113.00	168.0	222.00	277.0
bmi	24010.0	31.393328	7.876535	12.3	26.10	30.5	35.60	100.6
Year_last_admitted	13119.0	2003.892217	7.581521	1990.0	1997.00	2004.0	2010.00	2018.0
weight	25000.0	71.610480	9.325183	52.0	64.00	72.0	78.00	96.0
weight_change_in_last_one_year	25000.0	2.517960	1.690335	0.0	1.00	3.0	4.00	6.0
fat_percentage	25000.0	28.812280	8.632382	11.0	21.00	31.0	36.00	42.0
insurance_cost	25000.0	27147.407680	14323.691832	2468.0	16042.00	27148.0	37020.00	67870.0

Table6- Statistical Summary of the dataset

>Observations;

- * Applicants: 25,000 people, with IDs from 5,000 to 29,999.
- * Insurance Duration: Average of 4 years, ranging from 0 to 8 years.
- * Health Checkups: 77% had a checkup last year, and only 8% do adventure sports.
- * Doctor Visits: About 3 visits per year on average, with some up to 12.
- * Physical Activity: Average daily steps are 5,216, ranging from 2,034 to 11,255.
- * Age: Average age is 45, with a range from 16 to 74 years.
- * Health Conditions: 5% have heart disease; 10% have other major illnesses.
- * Glucose & BMI: Average glucose level is 168 mg/dL, and BMI is around 31 (high).
- * Hospital Admissions: For those with admissions, the last was around 2003 on average.
- * Weight: Average weight is 72 kg, ranging mostly between 64 and 78 kg.
- * Weight Change: Average change is 2.5 kg in the last year.
- * Fat Percentage: Average body fat is 29%, with a range from 11% to 42%.
- * Insurance Cost: Average cost is \$27,147, with a range from \$2,468 to \$67,870.

c) Understanding of attributes

Each attribute provides important information for assessing health risks:

- Years of Insurance: Shows loyalty or stability with the insurer.
- Regular Checkup and Doctor Visits: Indicate proactive health management.
- Adventure Sports: Higher-risk activities that might increase insurance cost.
- Occupation: Some jobs may impact lifestyle and health.
- Health Metrics (cholesterol, glucose, BMI): Direct indicators of health status.
- Medical History (heart disease, major surgeries): Key predictors of health risk.
- Lifestyle Factors (smoking, alcohol, exercise): Strongly linked to health outcomes.
- Weight, Weight Change, Fat Percentage: Help assess potential health risks.
- Insurance Cost: The target variable, showing the insurance cost for each person

3. Exploratory Data Analysis

a) Univariate Analysis

- Boxplots for all numerical columns

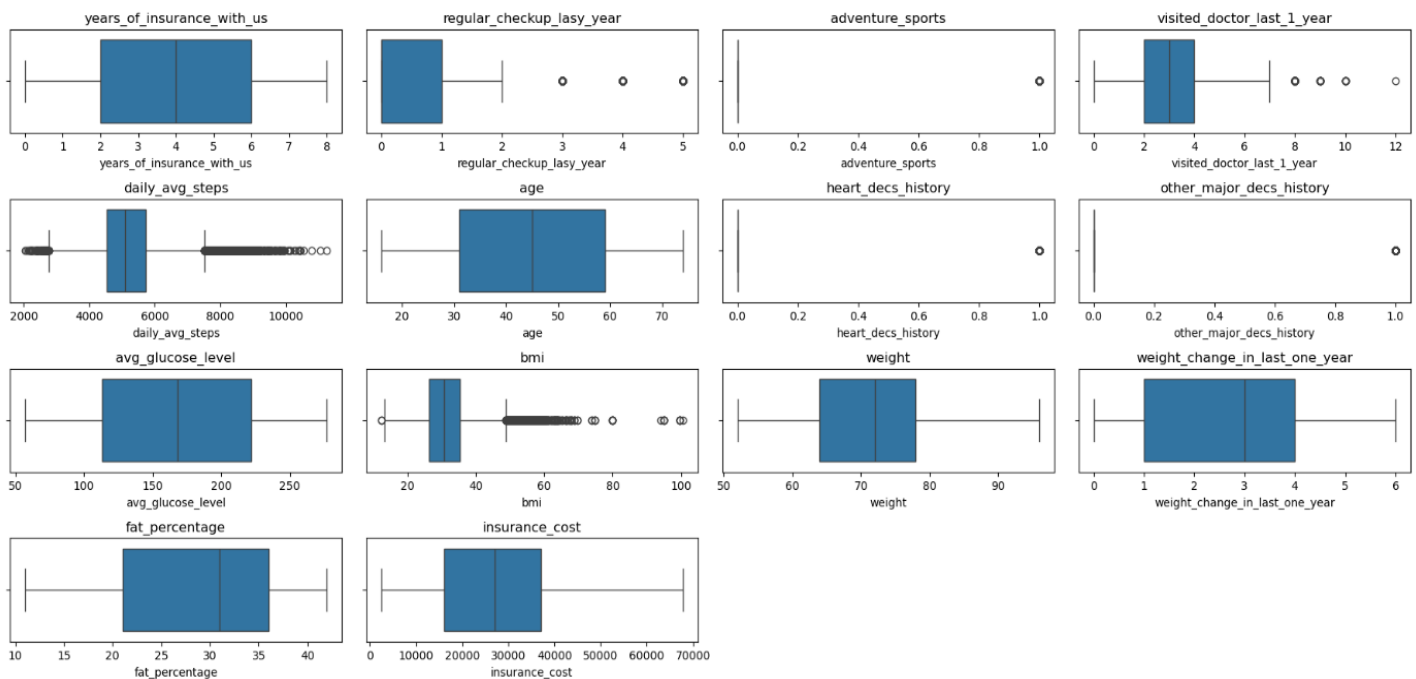


Figure1- Boxplots for all numerical variables

>Observations;

- * In the column years_of_insurance_with_us most people have lower years with insurance and a few have high values.
- * In the column regular_checkup_last_year most people had fewer checkups, with a few having more than average.
- * In the column adventure_sports this looks like a yes-or-no (0 or 1) variable.
- * In the column visited_doctor_last_year most people visited the doctor a few times, but some visited much more often.
- * In the column daily_avg_steps most people have an average of 4000-8000 steps daily, but some have very low or very high values.
- * In the column age are spread out evenly, with no major outliers.
- * In the column heart_decs_history and other_major_decs_history these seem to be yes-or-no variables for health conditions.
- * In the column avg_glucose_level most people have a lower glucose level, with a few high values.
- * In the column bmi has a few high outliers, but most values are below that range.
- * In the column Weight are spread fairly evenly with no major outliers.
- * In the column weight_change_in_last_one_year weight changes are spread evenly with no major outliers.
- * In the column fat_percentage vary widely with no big outliers.
- * In the column insurance_cost vary a lot, with no significant outliers.

- Distributions of all Continuous Columns

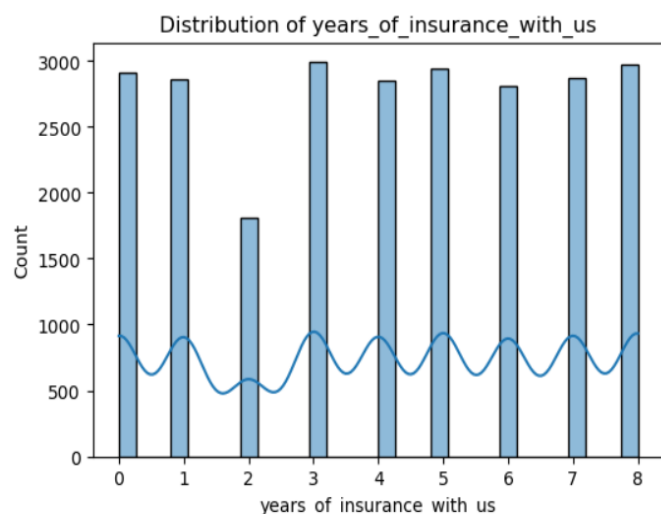


Figure2- Distributions of years of insurance with us

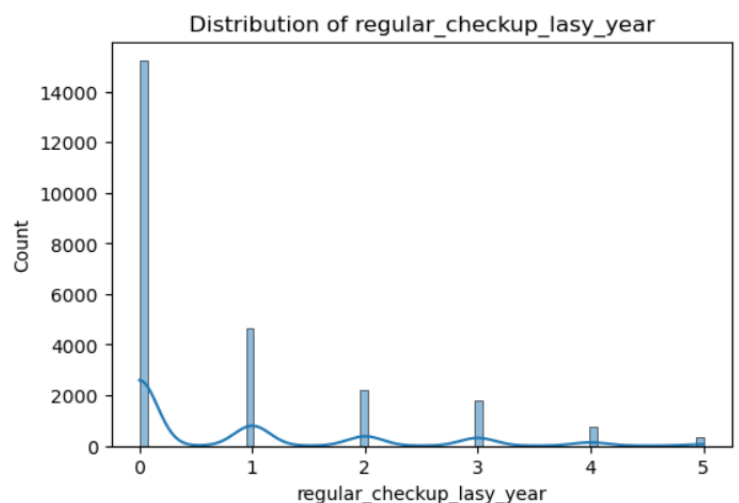


Figure3- Distribution of regular checkup last year

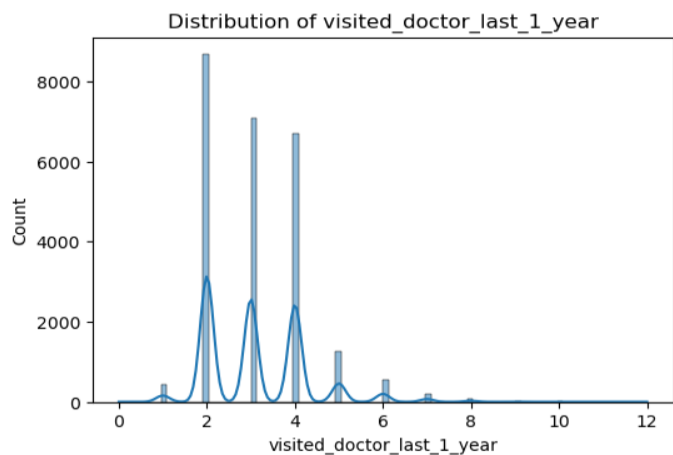


Figure4- Distribution of visited doctor last 1 year

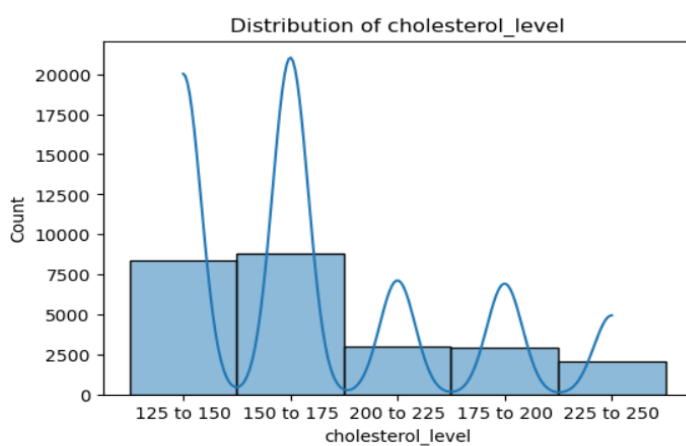


Figure5- Distribution of cholesterol level

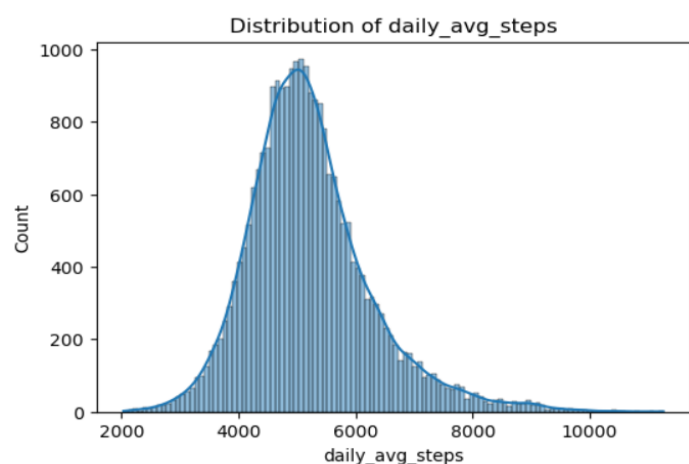


Figure6- Distributions of daily avg steps

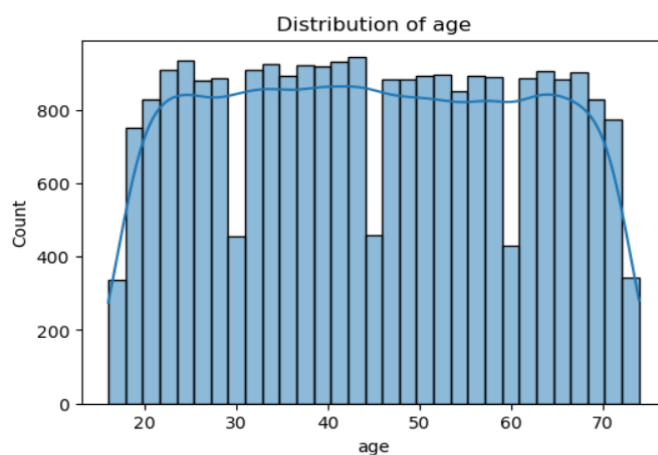


Figure7- Distribution of Age

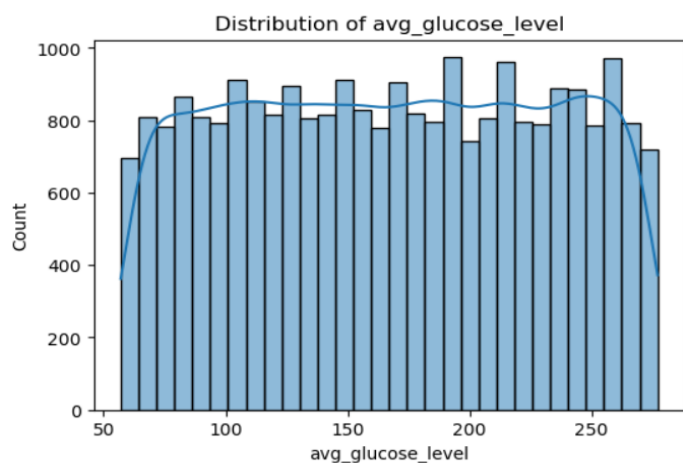


Figure8- Distribution of avg glucose level

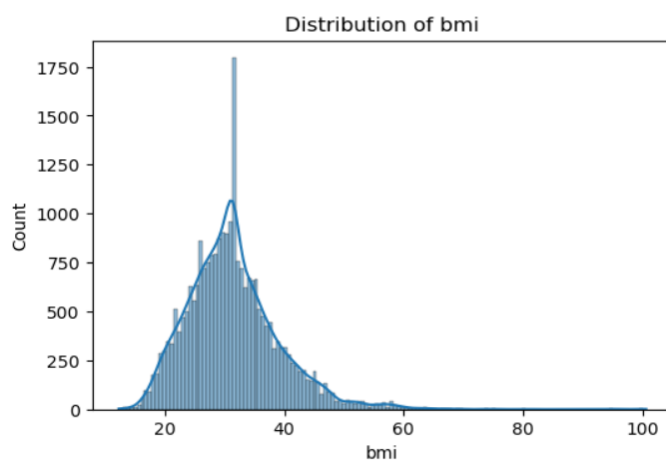


Figure9- Distribution of BMI

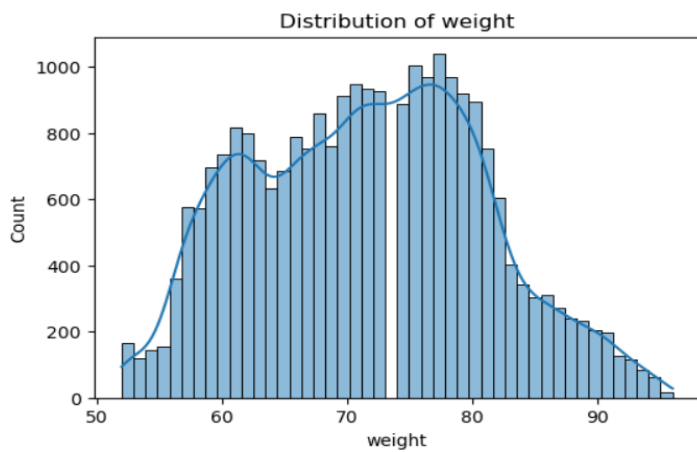


Figure10- Distribution of Weight

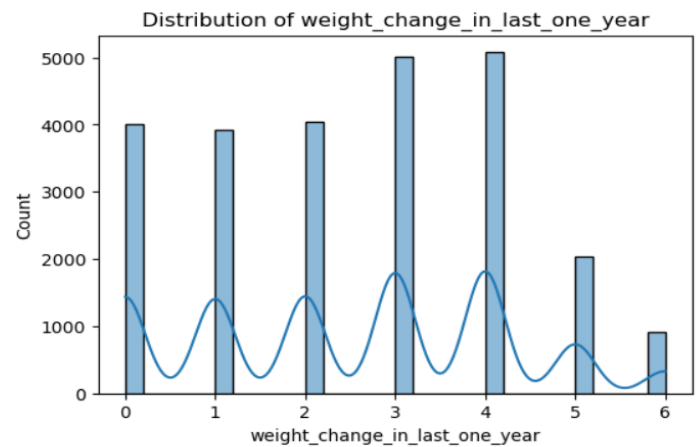


Figure11- Distribution of weight change in last one year

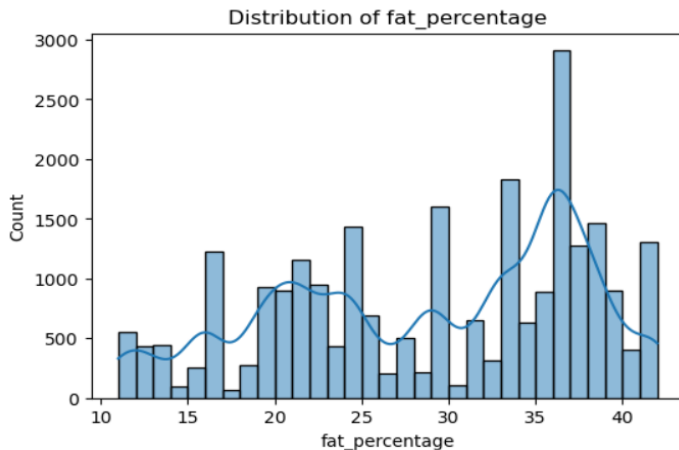


Figure12- Distribution of fat percentage

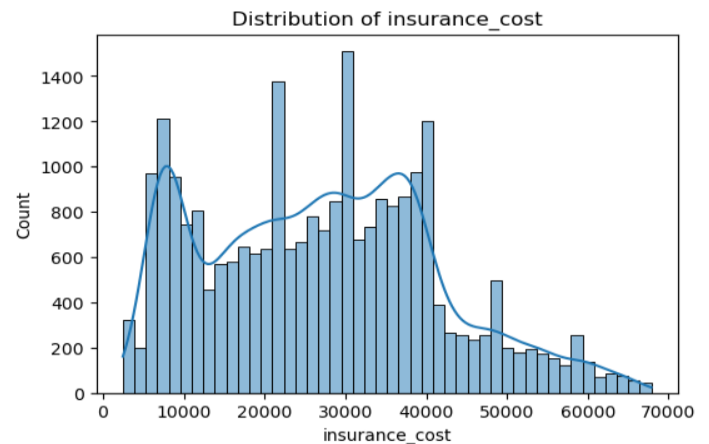


Figure13- Distribution of insurance cost

>Observations;

* We see in the column Years of Insurance with Us:

- For the right-skewed we can say most customers are new.
- For the left-skewed we can say many customers have been with us long-term.

* We see in the column Regular Checkup Last Year:

- Peaks of the bins around 0 or 1 suggested that people either consistently had checkups or didn't.
- Balanced of the bins suggested that a mix of people who did and didn't get checkups.

* We see in the column Visited Doctor Last 1 Year:

- For the right-skewed we can say most visited the doctor often.
- For the left-skewed we can say few visited the doctor.

* We see in the column Cholesterol Level:

- Normal bins shows healthy cholesterol levels.
- For the right-skewed we can say many have high cholesterol.

* We see in the column Daily Average Steps:

- Peak at low values: Many people are inactive.
- Right-skewed or balanced: Higher levels of physical activity.

* We see in the column Age:

- For the right-skewed we can say mostly younger people.
- For the left-skewed we can say mostly older people.

* We see in the column Average Glucose Level:

- Normal distribution of the bins suggested a healthy glucose levels.
- Skewed bins suggested that the glucose levels vary by health status.

* We see in the column BMI:

- For the right-skewed we can say many people are overweight.

* We see in the column Weight:

- Likely to match BMI trends, depending on population.

* We see in the column Weight Change in Last Year:

- Peaks of the bins around 0 indicate stable weight.
- For the help of positive/negative values we can understand people gained or lost weight.

* We see in the column Fat Percentage:

- For the right-skewed we can say many people have higher fat percentages.

* We see in the column Insurance Cost:

- For the right-skewed we can see that a few high-cost cases drive up average cost.

- Distributions of all Categorical Columns

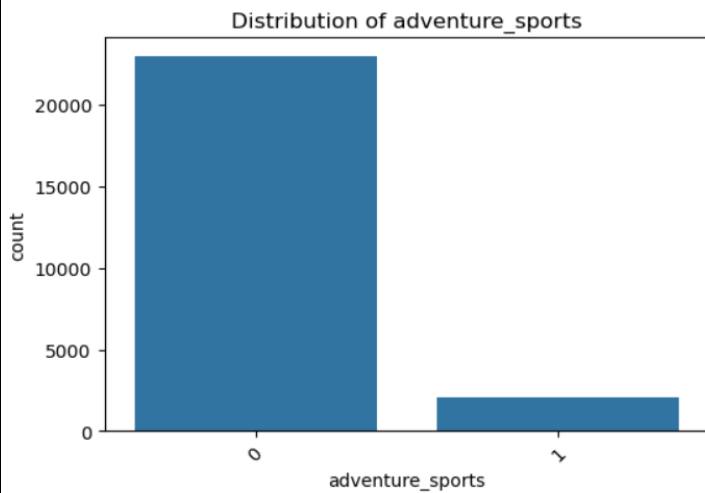


Figure14- distribution of adventure sports

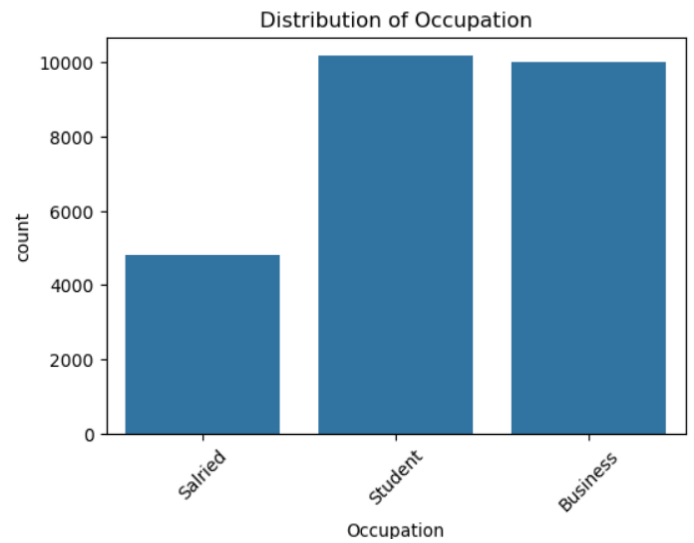


Figure15- Distribution of Occupation

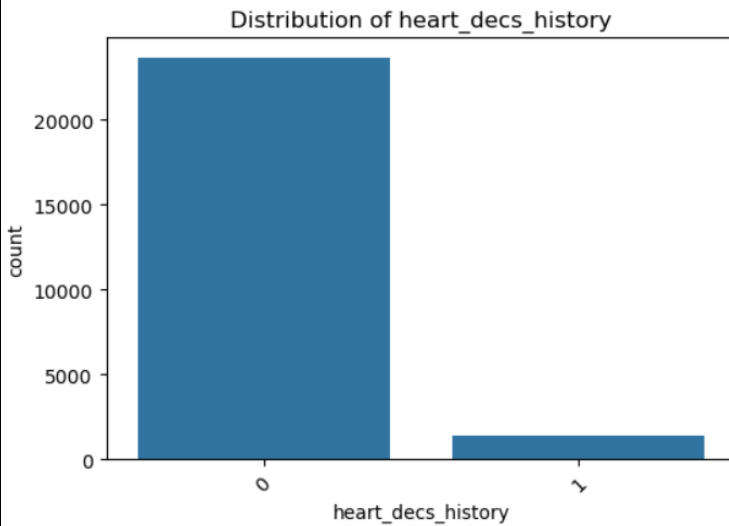


Figure16- Distribution of heart decs history

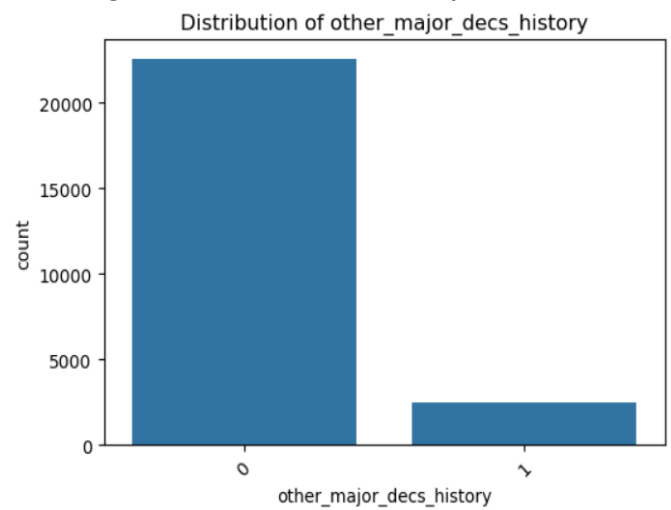


Figure17- Distribution of other major decs history

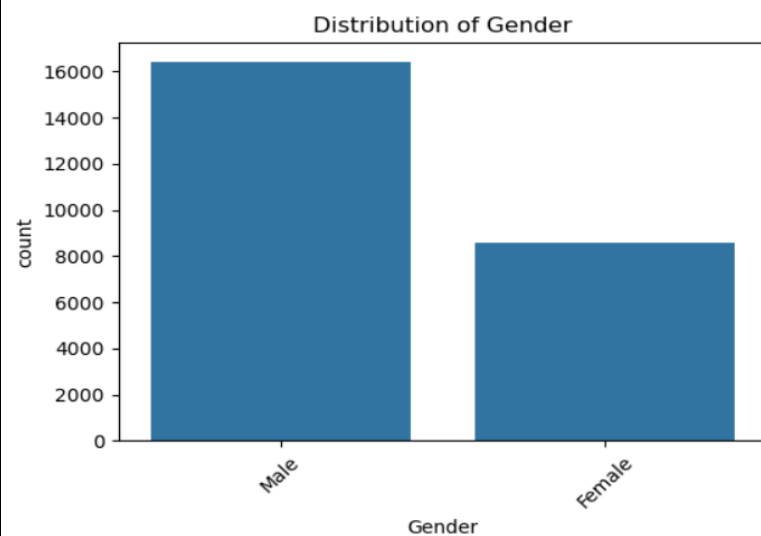


Figure18- Distribution of gender

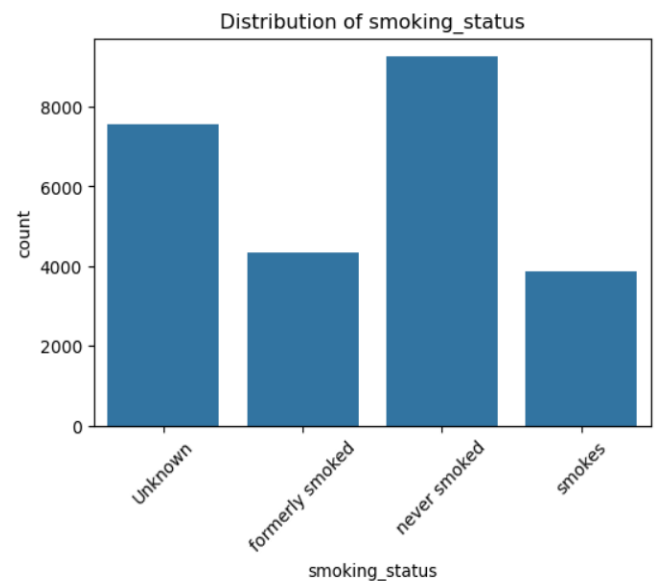


Figure19- Distribution of smoking status

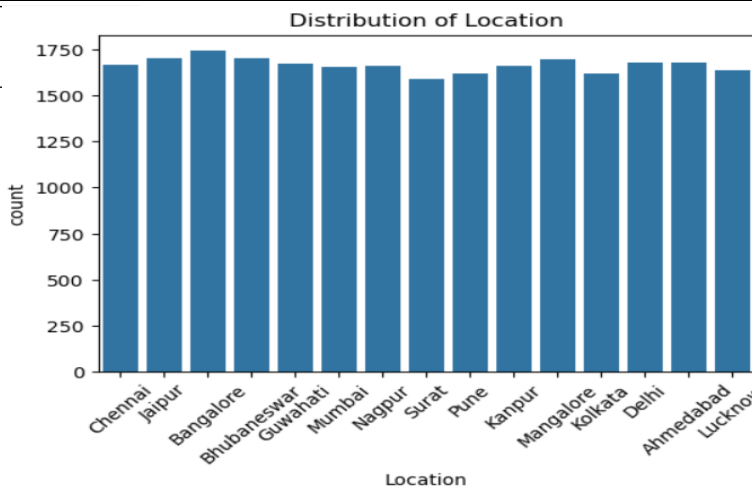


Figure20- Distribution of location

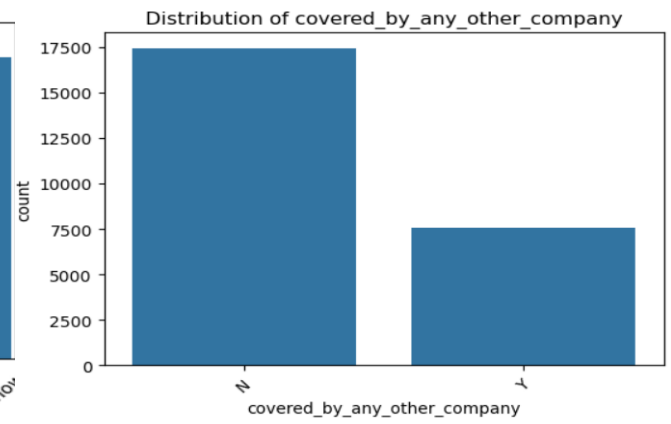


Figure21- Distribution of covered by any other company

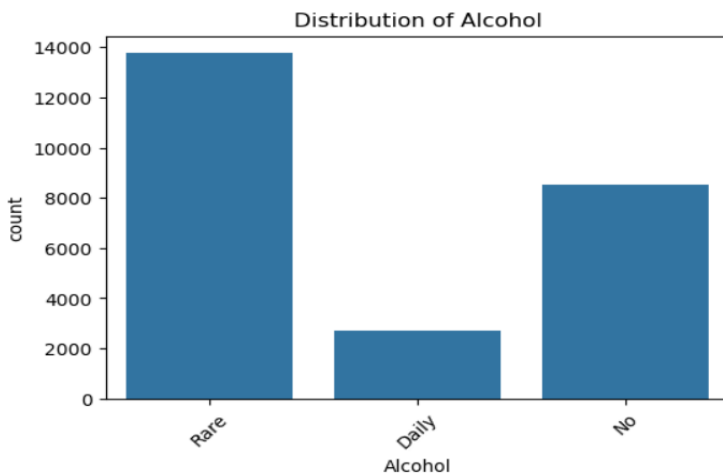


Figure22- Distribution of alcohol

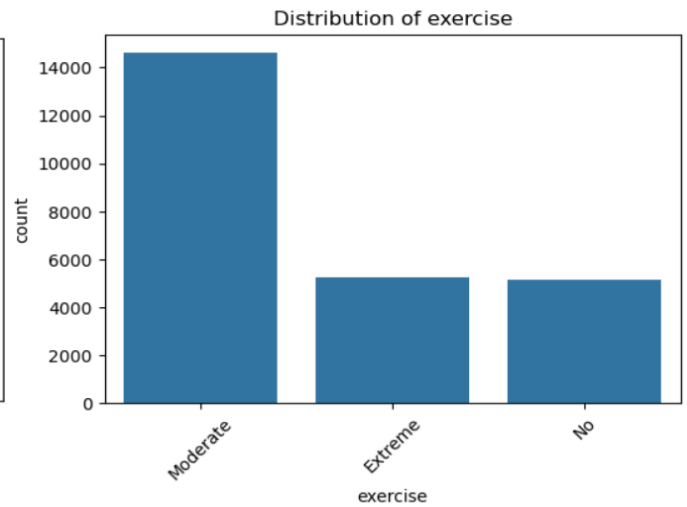


Figure23- Distribution of exercise

>Observations;

- Most of the customer are not involve in adventure sports.
- Occupationally students are more than business and salried.
- Most of the customer have no heart decs history.
- Most of the customer have no other major decs history.
- Male customers are more than female customers.
- Never smoked customer are higher than unknown, formerly smoke and smoked, smoked persons are very compared to other lower than 400.
- Approximately all location had same number of hospitals.
- Most of the customer had not covered by any other insurance company. Bellow 7500 customers are covered by other insurance company.
- Daily alcohol consumption customers are very lower than the never and rare, rare alcohol consumption customers are higher.
- Customers who are moderate in exercise are very higher than extreme and never do exercise customers.

b) Bivariate Analysis

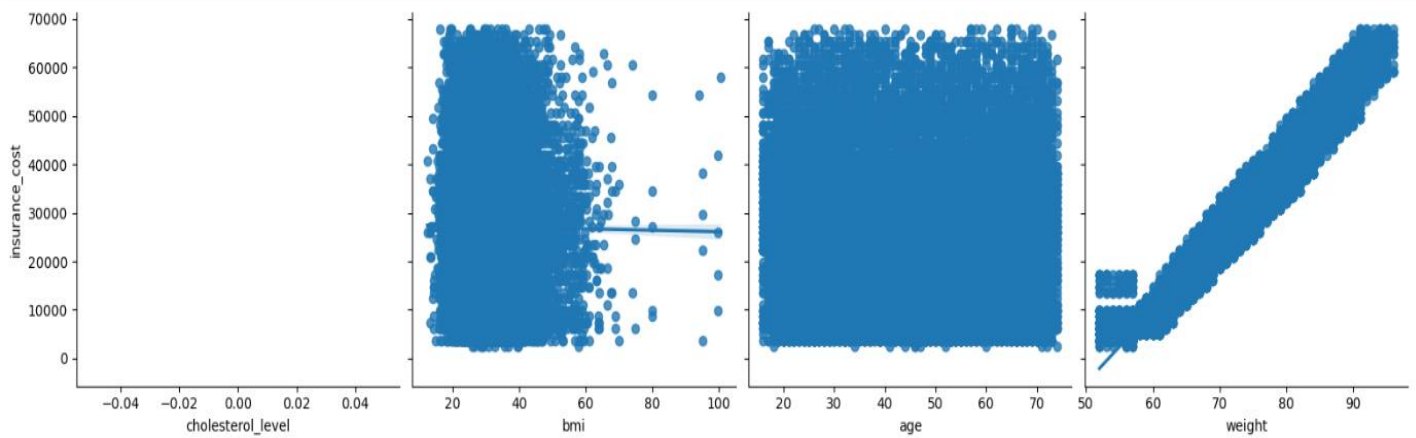


Figure24- Insurance cost vs cholesterol level, bmi, age, weight

>Observations;

- The scatter plot with regression for BMI likely shows a positive correlation as well.
- A clear upward trend would indicate that as BMI increases, insurance costs tend to rise.
- Age is often positively correlated with insurance cost due to the increased risk of health issues with aging.
- A strong positive trend here would suggest that older individuals tend to have higher insurance costs.
- The trend in the weight plot will help indicate there is a positive correlation between weight and insurance costs.
- A positive correlation would imply that higher weights might be associated with higher insurance costs.

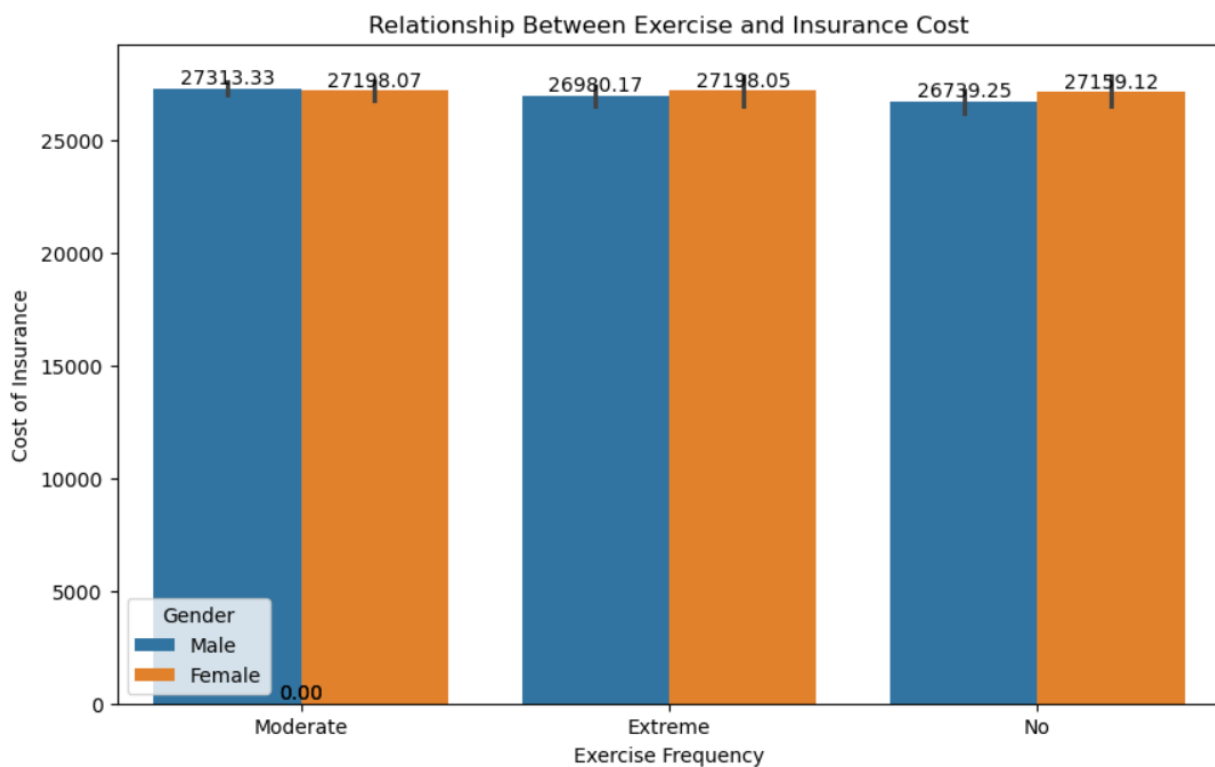


Figure25- Relationship Between Exercise and Insurance Cost

>Observations; Female customers who have never do exercise they have more chance for health risk and more chance to take insurance.

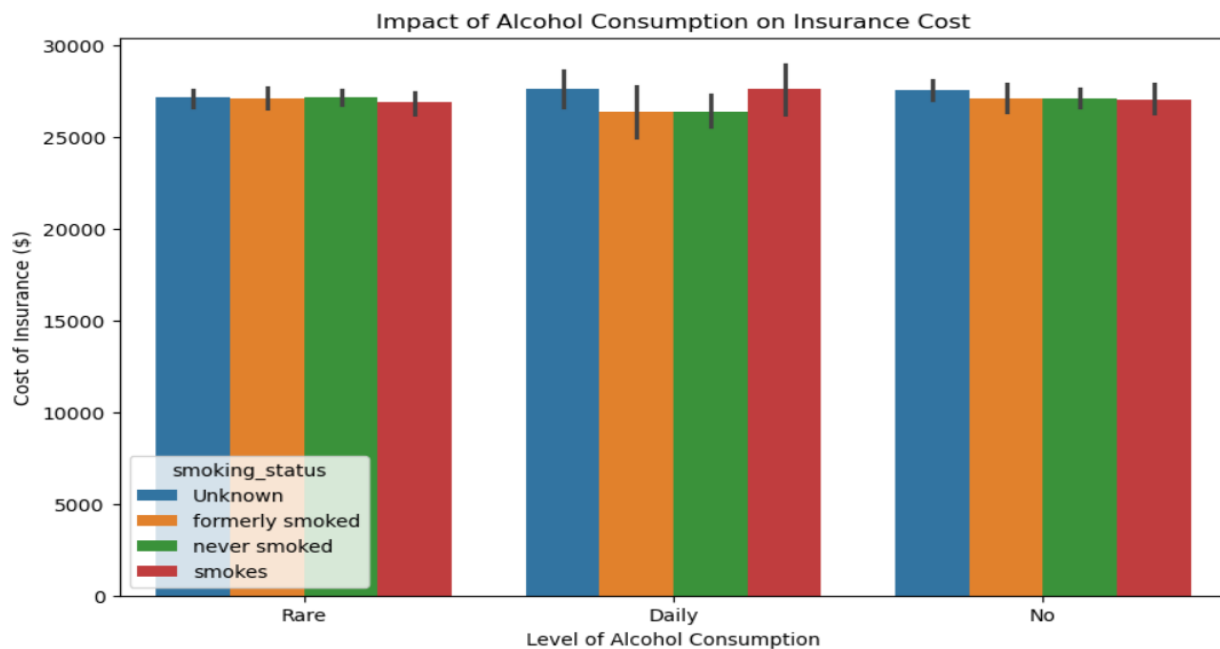


Figure26- Impact of Alcohol Consumption on Insurance Cost

>Observations; Customers who had consuming alcohol daily and also smoke has the chance of health risk more they use more insurance policies.

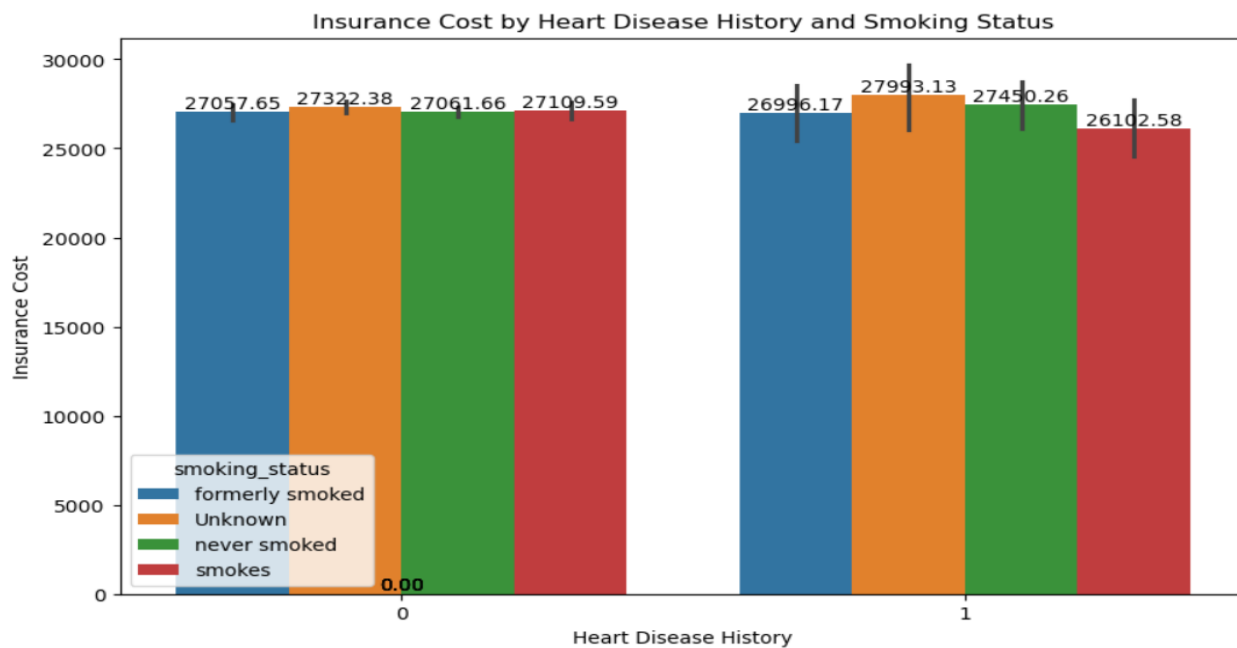


Figure27- Insurance cost by heart disease and smoking status

>Observations; Customers with no heart disease history but smoked is lower than customers who have no heart disease history but smoked is higher.

4. Business insights from EDA

- a) Is the data unbalanced? If so, what can be done? Please explain in the context of the business
b) Any business insights using clustering (if applicable) c) Any other business insights

-Check for data imbalance

Value Counts for Categorical Variables:

Gender

Male 16422

Female 8578

Name: count, dtype: int64

smoking_status

never smoked 9249

Unknown 7555

formerly smoked 4329

smokes 3867

Name: count, dtype: int64

Location

Bangalore 1742

Jaipur 1706

Bhubaneswar 1704

Mangalore 1697

Delhi 1680

Ahmedabad 1677

Guwahati 1672

Chennai 1669

Kanpur 1664

Nagpur 1663

Mumbai 1658

Lucknow 1637

Pune 1622

Kolkata 1620

Surat 1589

Name: count, dtype: int64

>Observations;

* Gender:

- There are nearly twice as many males (16,422) as females (8,578) in the dataset, which may affect gender-based analyses.

* Smoking Status:

- Most customers around 9,249 have never smoked.
- A large portion around 7,555 has an "Unknown" smoking status, which could complicate analysis.
- There's a similar count for those who currently smoke around 3,867 and those who formerly smoked around 4,329.

* Location:

- The data is spread across multiple cities, with Bangalore having the most entries around 1,742 and Surat the fewest nearly 1,589.

-Clustering on numerical variables only

- Features like age, BMI, and insurance cost have different ranges, which can affect clustering results. Without scaling, features with larger ranges could dominate the clustering process, giving skewed results.
- Using StandardScaler changes each feature to have a mean of 0 and a standard deviation of 1. This puts all features on the same scale, making them easier to compare directly.
- After scaling, the data values are now in standard deviations from the mean.
- We are using KMeans clustering for these three numerical clusters ('age', 'bmi', 'insurance_cost')

- Visualizing clusters

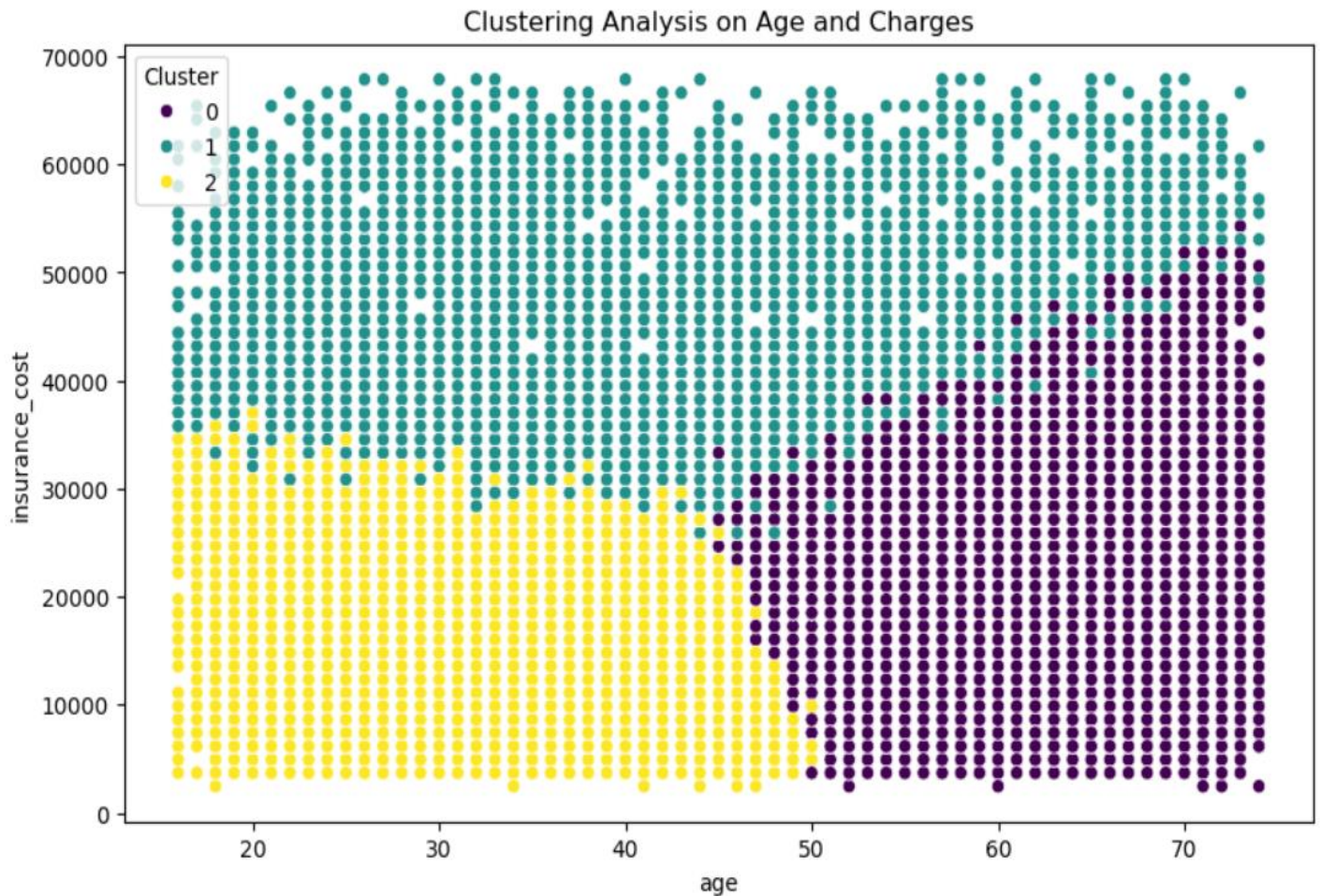


Figure28- Clustering Analysis on Age and Insurance cost

>Observations; This scatter plot shows a clustering of people based on their age and insurance costs, divided into three groups which is called clusters and it is also shown in different colors.

Cluster 0 which color is Purple:

- Mostly older customers around age 50 and above 50.
- It shows that they have a wide range of insurance costs, often in the mid to high range.

Cluster 1 which color is Teal:

- It includes customers of all ages.
- Most of the customer have medium to high insurance costs.
- This cluster 1 may have higher insurance needs, possibly due to lifestyle or health issues.

Cluster 2 which color is Yellow:

- Mainly younger customer under 40.
- It shows that they have lower insurance costs.
- Likely represents younger people with fewer health needs.

Overall Patterns from those clusters:

- Older customers are mostly in the clusters with higher insurance costs that is clusters 0 and 1.
- Younger customers are generally in the low-cost cluster that is cluster 2.

-Further insights may involve analyzing the differences in charges between smokers and non-smokers, as well as identifying regions associated with higher charges

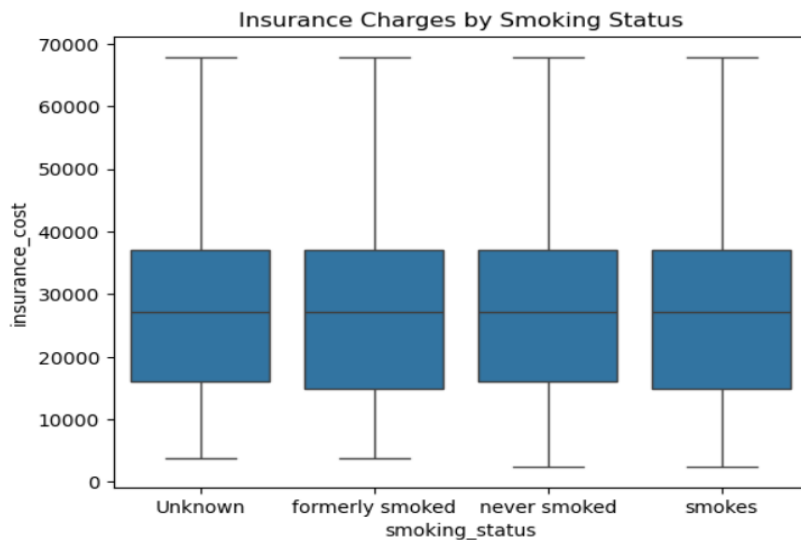


Figure29- Insurance charges by smoking status

>Observations;

- Insurance costs are similar for all smoking statuses like Unknown, formerly smoked, never smoked, and smokes.
- The box plots show there is a big range in insurance costs within each group.
- Smoking status doesn't seem to have a big impact on insurance costs in this data.

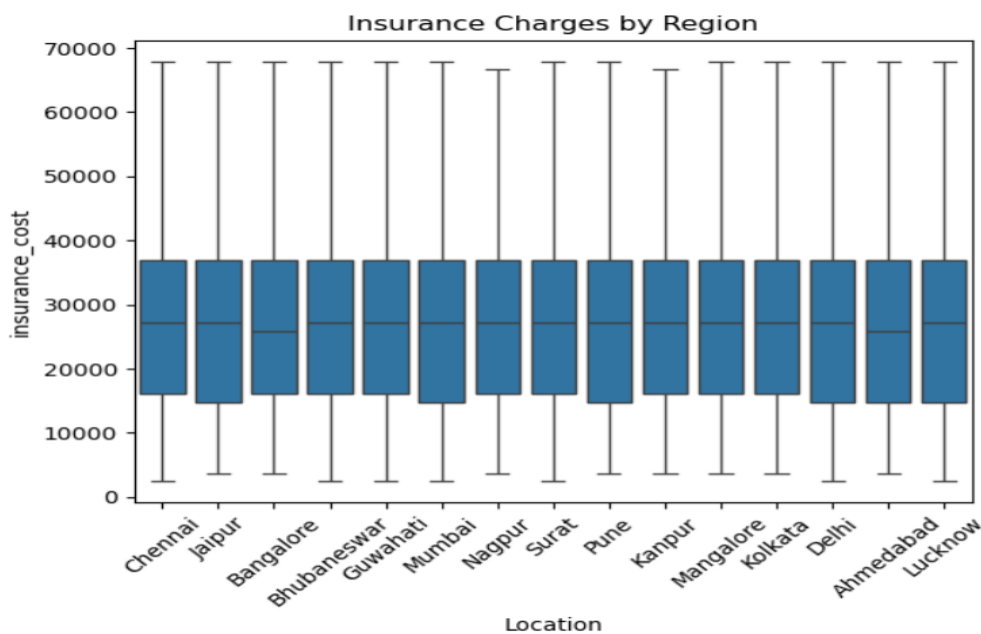


Figure30- Insurance charges by region

>Observations;

- Insurance costs appear similar across all locations, with no region showing notably higher or lower costs.
- Every region has a broad range of insurance costs, as shown by the long whiskers.
- Location doesn't seem to have a major impact on insurance costs in this data.

5. EDA (Exploratory Data Analysis) and Business Implications

a. Implications from Univariate Analysis

Understanding individual variables and their implications:

- Focus on retaining new customers with loyalty programs and discounts.
- Encourage health checkups by offering incentives to reduce future claims.
- Market adventure sports coverage to a small, niche audience.
- Target high-visit customers with wellness programs to manage health risks and costs.
- Create plans tailored to young and older customers' needs.
- Offer weight management programs and adjust premiums for those at higher risk.
- Implement pricing strategies based on individual risk levels.

b. Implications from Bivariate Analysis

Examining relationships between two variables:

- BMI vs. Insurance Cost Implication: Motivate healthy lifestyles to reduce costs and premiums.
- Age vs. Insurance Cost Implication: Adjust premiums for older customers and offer loyalty benefits to retain them.
- Daily Average Steps vs. Insurance Cost Implication: Promote fitness programs and reward active lifestyles.
- Smoking vs. Insurance Cost Implication: Provide smoking cessation programs to lower health risks and expenses.

c. Multivariate Analysis

Understanding complex patterns and relationships:

- Customer Clusters by Age and Insurance Cost:
 - Implication: Focus on low-cost, high-volume plans for this group.
- Alcohol and Smoking Combined Implication: Charge higher premiums or provide targeted health intervention programs.

Overall Business Implications from EDA

- ✓ Customer Segmentation: Use clusters and patterns to design insurance plans for specific customer groups.
- ✓ Risk Management: Identify high-risk individuals to adjust pricing and offer targeted wellness programs.
- ✓ Preventive Measures: Promote checkups, fitness, and smoking cessation programs to reduce claims and costs.
- ✓ Profit Optimization: Use pricing strategies to manage high-cost customers and maintain profitability.
- ✓ Customer Retention: Develop loyalty programs to keep new customers and reward long-term ones.

6. Data Cleaning and Pre-processing

>Outlier Treatment;

* Presence of Outliers:

- The columns which are regular_checkup_last_year, adventure_sports, visited_doctor_last_1_year, daily_avg_steps, heart_decs_history, other_major_decs_history, bmi were identified as containing potential outliers.
- These columns likely represent numerical or continuous variables prone to extreme values that could distort data analysis or model training.

* IQR Thresholding:

- Outliers are defined based on the Interquartile Range (IQR) method:
 - Any value below $Q1 - 1.5 * IQR$ or above $Q3 + 1.5 * IQR$ is considered an outlier.
- This rule is robust for detecting extreme values in data that might not follow a normal distribution.

* Capping Outliers:

- Values exceeding the lower or upper bounds were not removed but replaced with the respective threshold.

* Approach

- Identifying Outliers:
 - The IQR method was used for its simplicity and effectiveness in identifying outliers without making assumptions about the underlying data distribution.
- Treatment of Outliers:
 - Capping:
 - Instead of removing or discarding outliers, they were capped to the respective threshold values.
 - This ensures that extreme values are brought within the range of the bulk of the data.

- Iterative Application:
 - The process was applied iteratively to multiple columns, ensuring uniform treatment of all outlier-prone variables.

>Handling Missing Values;

* Approach:

- Mean Imputation for the bmi Column: Missing values in the bmi column were replaced with the column's mean value.

* Reasoning:

- Imputation using the mean is simple and effective when the missing data is limited and the distribution is not heavily skewed.
- This ensures that the missing values do not distort statistical relationships or introduce bias into the model.

>Encoding Categorical Variables

* Ordinal Columns:

- Label Encoding: Ordinal columns like smoking_status and cholesterol_level were label-encoded.
 - This assigns a numerical value to each category based on the natural order of the categories (e.g., low, medium, high).

* Reasoning:

- Ordinal data has an inherent order, so encoding the categories with numbers preserves this order.

* Nominal Columns:

- One-Hot Encoding: Nominal columns like 'Occupation', 'Gender', 'Location', 'covered_by_any_other_company', 'Alcohol', 'exercise' are one-hot encoded.
 - Each category was converted into a binary column.

* Reasoning:

- Nominal data has no inherent order, so one-hot encoding ensures equal treatment of all categories without imposing an arbitrary order.
- The use of meaningful names in one-hot encoding improves interpretability of the transformed data.

>Why This Approach Was Chosen

- **Outlier Treatment:**
 - Capping minimizes distortion of the dataset while addressing the extreme values that could bias statistical models.
- **Missing Values:**
 - Mean imputation is computationally simple and maintains dataset size, particularly suitable when the missingness is not systematic.
- **Encoding Categorical Variables:**
 - The chosen methods ensure numerical representations of categorical variables that align with their data types.
 - Both techniques facilitate compatibility with machine learning algorithms, which generally do not handle raw categorical data.

7. Model building and interpretation.

Label Encoding for ordinal columns these are 'smoking_status', 'cholesterol_level'.

One-Hot Encoding for nominal columns with meaningful names these are 'Occupation', 'Gender', 'Location', 'covered_by_any_other_company', 'Alcohol', 'exercise'.

	years_of_insurance_with_us	regular_checkup_lasy_year	adventure_sports	visited_doctor_last_1_year	cholesterol_level	daily_avg_steps	age	heart_decs_history	oth
0	3	1.0	0.0	2.0	0	4866.0	28	0.0	
1	0	0.0	0.0	4.0	0	6411.0	50	0.0	
2	1	0.0	0.0	4.0	0	4509.0	68	0.0	
3	7	2.5	0.0	2.0	0	6214.0	51	0.0	
4	3	1.0	0.0	2.0	0	4938.0	44	0.0	
...	...	...	...	...	...	...	...	...	...
24995	3	0.0	0.0	4.0	0	5614.0	22	0.0	
24996	6	0.0	0.0	4.0	0	4719.0	58	0.0	
24997	7	0.0	0.0	2.0	0	5624.0	34	0.0	
24998	1	0.0	0.0	2.0	0	7510.5	27	0.0	
24999	8	2.0	0.0	4.0	0	5882.0	22	0.0	

25000 rows × 45 columns

other_major_decs_history	avg_glucose_level	bmi	smoking_status	weight	weight_change_in_last_one_year	fat_percentage	insurance_cost	Cluster	Occupation_Busi
0.0	97	31.200000	0	67		1	25	20978	2
0.0	212	34.200000	1	58		3	27	6170	2
0.0	166	40.400000	1	73		0	32	28382	0
0.0	109	22.900000	0	71		3	37	27148	0
0.0	118	26.500000	2	74		0	34	29616	1
...	...	...	...	...		...	...	...	...
0.0	145	36.100000	3	79		4	40	39488	1
0.0	134	31.300000	2	66		2	28	14808	0
0.0	151	31.393328	0	76		1	35	33318	1
0.0	66	26.600000	0	74		0	40	30850	2
0.0	245	41.600000	1	57		4	21	6170	2

nder_Female	Gender_Male	Location_Ahmedabad	Location_Bangalore	Location_Bhubaneswar	Location_Chennai	Location_Delhi	Location_Guwahati	Location_Jaipur	Loca
False	True	False	False	False	True	False	False	False	
False	True	False	False	False	False	False	False	True	
True	False	False	False	False	False	False	False	True	
True	False	False	False	False	True	False	False	False	
False	True	False	True	False	False	False	False	False	
...	...	...	...	...	...	...	...	...	...
False	True	False	False	False	False	False	False	False	
False	True	False	False	False	False	False	False	False	
False	True	False	False	True	False	False	False	False	
False	True	False	False	False	False	False	False	False	
False	True	False	False	False	True	False	False	False	

Table7- Dataset after encoding the Nominal and Ordinal columns

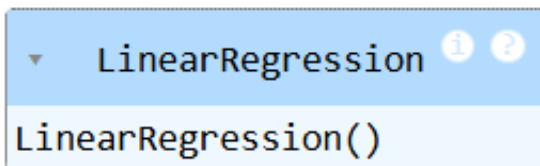
- Scaling the Dataset

	years_of_insurance_with_us	regular_checkup_lasy_year	adventure_sports	visited_doctor_last_1_year	cholesterol_level	daily_avg_steps	age	heart_decs_history	oti
0	-1.572831	1.471346	0.0	0.815534	0.0	0.066506	0.003132	0.0	
1	0.728712	1.471346	0.0	-0.081953	0.0	2.409803	0.934323	0.0	
2	0.728712	-0.710966	0.0	0.815534	0.0	0.669526	-0.245185	0.0	
3	0.728712	0.380190	0.0	-0.081953	0.0	-0.742347	0.623926	0.0	
4	-0.038469	2.016924	0.0	-0.081953	0.0	2.409803	-1.238456	0.0	

Table8- Head of the X_train scaled

Metric of Choice

- Linear Regression Model



-Predictions and Evaluation of Linear Regression

Linear Regression:

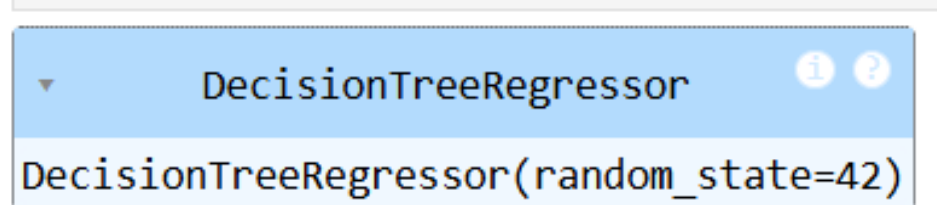
MAE: 2709.212432579549

MSE: 11127798.236842277

R2 Score: 0.9449333199569503

MAPE: 15.08067779351285

-Decision Tree Regression



-Predictions and Evaluation of Decision Tree Regression

Decision Tree:

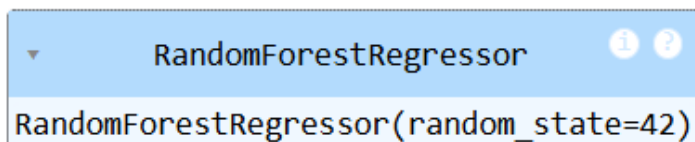
MAE: 3201.3250666666668

MSE: 17181357.465066668

R2 Score: 0.9149768629789096

MAPE: 15.865327564666348

-Random Forest Model



-Predictions and Evaluation of Random Forest Model

Random Forest:

MAE: 2354.444029333333

MSE: 8775528.458872693

R2 Score: 0.9565736898200135

MAPE: 11.799753147926316

-Observations from all the models;

1. Linear Regression:
 - Good performance with an R^2 of 0.9449, meaning it explains most of the variance in insurance costs.
 - MAE, MSE and MAPE are higher than Random Forest, showing it struggles with complex patterns.
2. Decision Tree:
 - Lowest performance with an R^2 of 0.9144.
 - Highest MAE, MSE and MAPE, indicating overfitting and poor generalization.
3. Random Forest:
 - Best performance with the lowest MAE (2354.63), MSE (8,765,928.44) and MAPE (11.799753147926316).
 - Highest R^2 (0.9566), showing it handles complex relationships well.

So, we can say that the Random Forest is the best model for predicting insurance costs, offering the highest accuracy and reliability.

Interpretation of the Model(s)

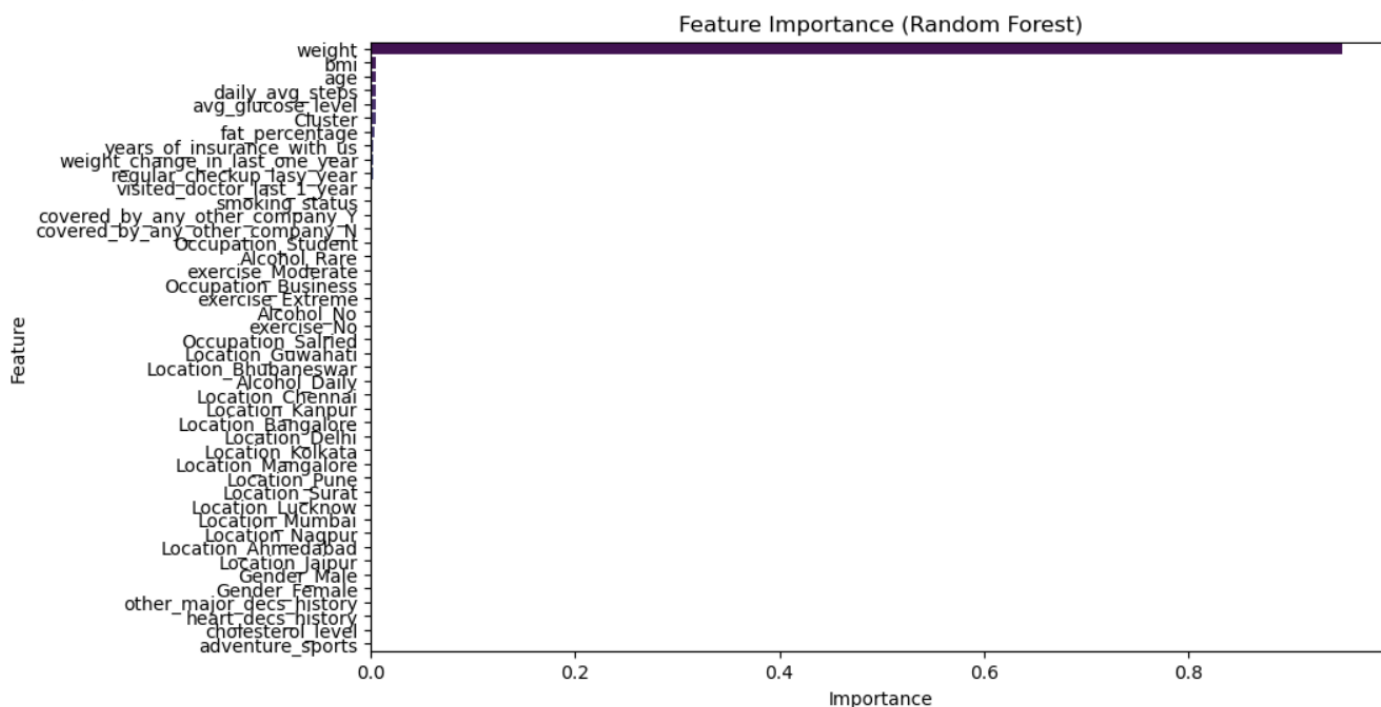


Figure 31- Feature Importance (Random Forest)

-Observations;

* Most Important Factors:

- Weight is the most critical factor affecting insurance costs.
- BMI, age, and daily steps are also very important for predicting costs.

* Lifestyle and Health History:

- Factors like smoking, exercise, and past diseases also have a strong impact on insurance costs.

* Less Important Factors:

- Location and occupation have little influence on the costs.

* Business Suggestions:

- Encourage weight management and healthy lifestyles to reduce insurance costs.
- Promote exercise programs to improve customer health and lower risks.

-Comparing true vs predicted values for the best model

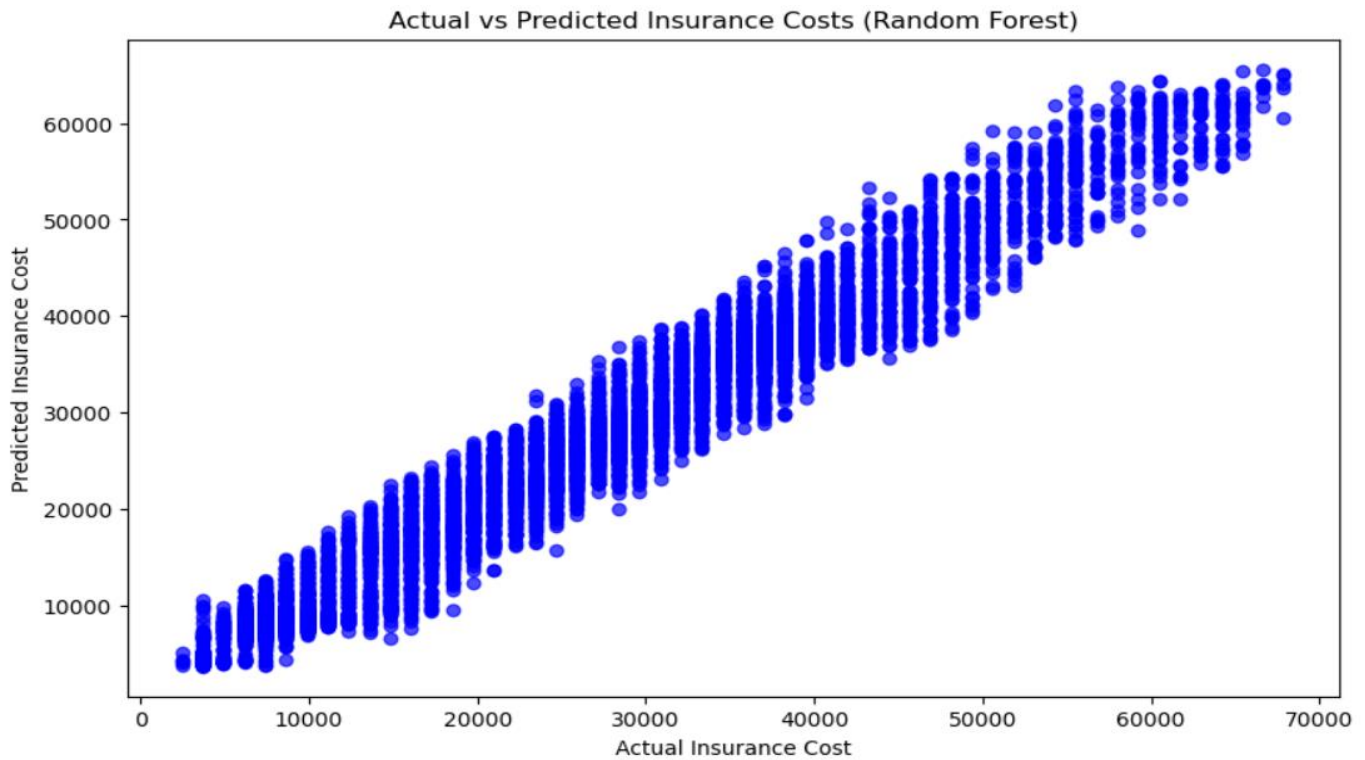


Figure 32- Actual vs Predicted Insurance Cost (Random Forest)

8. Model Tuning

Decision Tree Tuning

Fitting 3 folds for each of 36 candidates, totalling 108 fits

Tuned Decision Tree:

MAE: 2527.896765969707

MSE: 10015439.596313411

R2 Score: 0.950437903707249

MAPE: 15.08067779351285

Random Forest Tuning

Fitting 3 folds for each of 81 candidates, totalling 243 fits

Tuned Random Forest:

MAE: 2333.379623666326

MSE: 8574105.06049489

R2 Score: 0.9575704474530672

MAPE: 11.799753147926316

Summary Comparison

Model Comparison:

	Model	MAE	MSE	R2 Score	MAPE Score
0	Linear Regression	2709.210000	1.112779e+07	0.944900	15.080700
1	Decision Tree	3199.350000	1.728815e+07	0.914400	15.865300
2	Tuned Decision Tree	2527.896766	1.001544e+07	0.950438	11.799753
3	Random Forest	2354.630000	8.765928e+06	0.956600	11.799800
4	Tuned Random Forest	2333.379624	8.574105e+06	0.957570	11.799753

Table9- Model Comparison

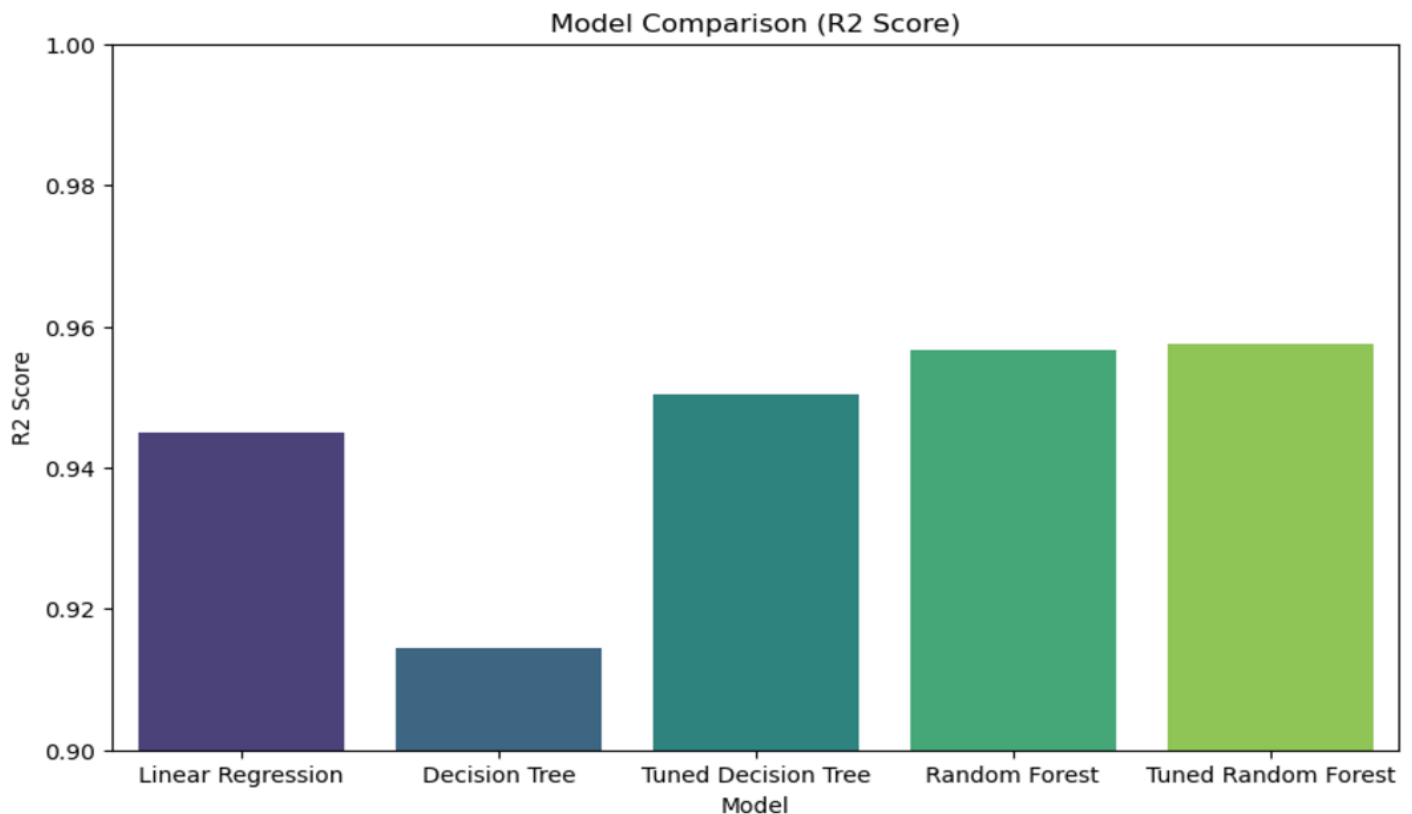


Figure 33- Model Comparison (R2 Score)

-Observations;

* Linear Regression:

- Good performance ($R^2 \sim 0.94$) but less effective than Random Forest.

* Decision Tree:

- Lowest performance ($R^2 \sim 0.91$); struggles to generalize well.

* Tuned Decision Tree:

- Better than the basic Decision Tree, with improved R^2 .

* Random Forest:

- Strong performance ($R^2 \sim 0.96$); handles complexity and generalizes well.

* Tuned Random Forest:

- Best model with the highest R^2 (~ 0.96); reliable and accurate.

The Tuned Random Forest model is the most reliable and precise, making it the optimal choice for accurately predicting insurance costs.

Best Model Interpretation - Actual vs Predicted for Tuned Random Forest

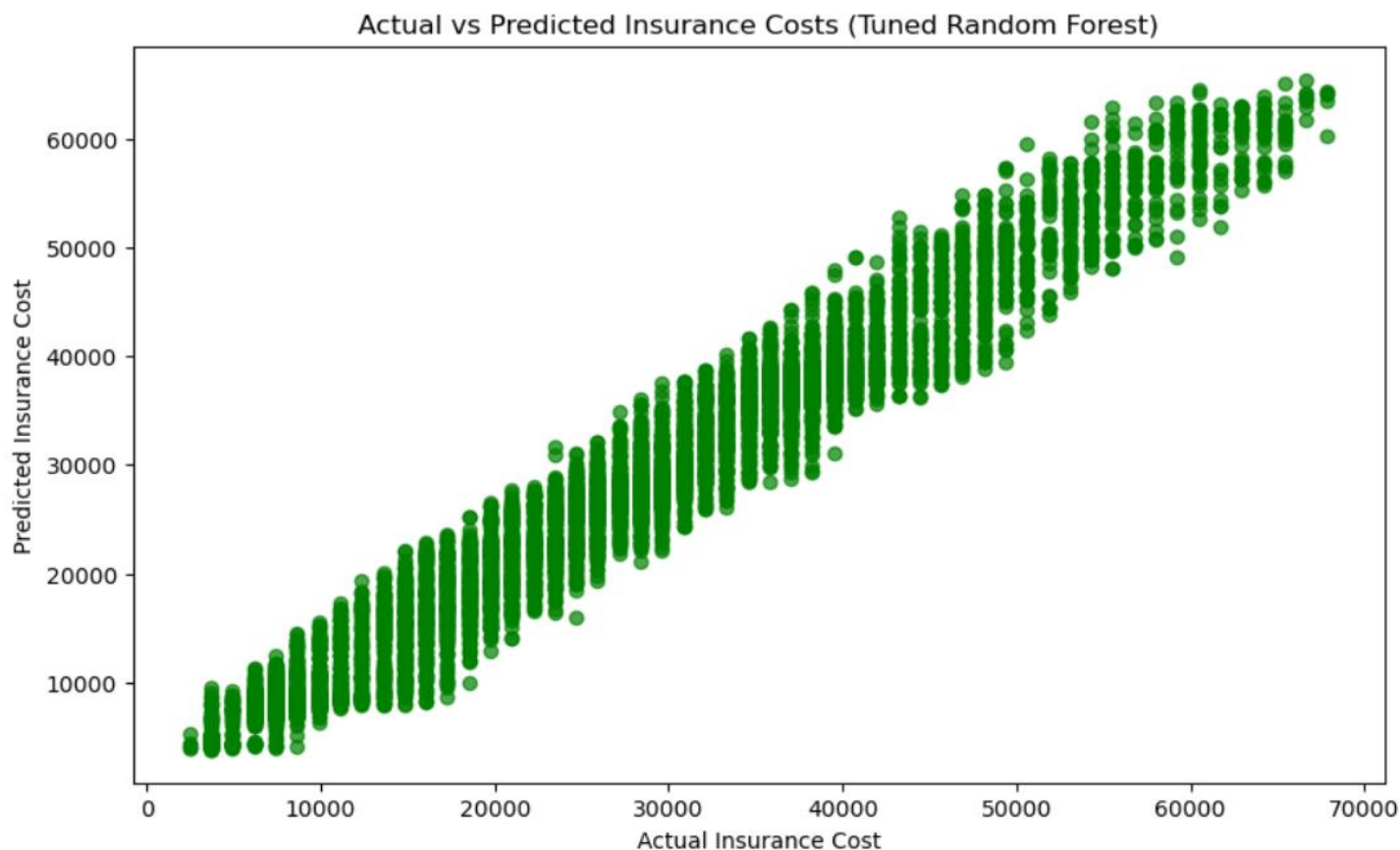


Figure 34- Actual vs Predicted Insurance Cost (Tuned Random Forest)

-Observations;

- **Strong Correlation:** The points are close to the diagonal, showing a strong relationship between actual and predicted insurance costs.
- **High Accuracy:** The points are tightly grouped near the line, meaning the model predicts accurately.
- **Minimal Deviation:** There is little spread from the line, indicating small prediction errors.
- **Reliable Performance:** The model works well for both low and high insurance costs.

Why was a particular model chosen?

- **Tuned Random Forest was chosen because:**
 - **Best Accuracy:** It has the highest R^2 Score is 0.9576, showing it explains the most variance in the data.
 - **Lowest Errors:**
 - MAE: 2333.38 (smallest among all models).
 - MSE: 8.57e+06 (lowest mean squared error).
 - **Consistency:** The Random Forest model, especially when tuned, performs better on both accuracy and error metrics compared to simpler models like Linear Regression and Decision Tree.
 - **Robustness:** It effectively handles non-linear relationships, interactions between variables, and reduces overfitting.

Efforts to Improve Model Performance:

- **Tuning Hyperparameters:**
 - The Decision Tree and Random Forest models were tuned using grid search or random search techniques to find the optimal parameters (e.g., number of trees, max depth, and minimum samples).
 - The Tuned Random Forest improved further compared to the default Random Forest.
- **Feature Engineering:**
 - Key features such as weight, BMI, age, and activity levels were carefully analyzed and retained due to their strong impact on predictions.

➤ **Comparing Models:**

- Linear Regression, Decision Tree, and Random Forest models were compared based on key metrics MAE, MSE, R^2 Score, and MAPE.
- The Tuned Random Forest was found to be the most accurate and reliable model for predicting insurance costs.

9- Final interpretation / Recommendation

>**Final interpretation**

Based on the analysis and model performance, the following conclusions can be drawn:

➤ **Key Factors Affecting Insurance Costs**

- Weight, BMI, daily steps, and cholesterol levels greatly influence insurance costs, showing the importance of a healthy lifestyle.
- Smoking, alcohol consumption, and exercise habits strongly affect costs, linking lifestyle to health risks.
- Older age and conditions like heart disease increase costs, highlighting the impact of health history.

➤ **Best Model for Prediction**

- The Tuned Random Forest model is the most accurate and reliable for predicting insurance costs.
- It has the lowest errors of MAE, MSE, and MAPE and the highest R^2 score (0.9576), meaning it explains 95.76% of the variation in costs.
- The model works well across all data, even for high-risk groups, making it a robust choice for predictions.

>**Recommendations**

- ✓ Support weight loss, exercise, and quit-smoking initiatives to lower premiums and improve health.
- ✓ Offer custom premiums based on health metrics to motivate healthier lifestyles.
- ✓ Provide discounts for tracking and improving BMI, steps, and activity levels.
- ✓ Apply the Tuned Random Forest model for fair and accurate premium pricing.
- ✓ Focus on smokers or inactive individuals with tailored plans to reduce claims.
- ✓ Educate customers on how lifestyle choices affect premiums to promote better decisions.